

5052 aluminium alloy

5052 is an aluminium–magnesium alloy, primarily alloyed with magnesium and chromium. 5052 is not a heat treatable aluminum alloy, but can be hardened through cold working.^[2]

Chemical properties

The alloy composition of 5052 is:^[2]

- Magnesium - 2.2%-2.8% by weight
- Chromium - 0.15%-0.35% maximum
- Copper - 0.1% maximum
- Iron - 0.4% maximum
- Manganese - 0.1% maximum
- Silicon - 0.25% maximum
- Zinc - 0.1% maximum
- Remainder Aluminium

A similar alloy A5652 exists differing only in impurities limits.

A5052	
Physical properties	
Density (<i>ρ</i>)	2.68 g/cm ³ ^[1]
Mechanical properties	
Young's modulus (<i>E</i>)	69.3 GPa (10,050 ksi)
Tensile strength (<i>σ_t</i>)	195–290 MPa (28.3–42.1 ksi)
Elongation (<i>ε</i>) at break	7-27%
Poisson's ratio (<i>ν</i>)	0.33
Thermal properties	
Melting temperature (<i>T_m</i>)	607 °C (1,125 °F)
Thermal conductivity (<i>k</i>)	138 W/m*K
Linear thermal expansion coefficient (<i>α</i>)	2.38*10 ^{−5} K ^{−1}
Specific heat capacity (<i>c</i>)	880 J/kg*K
Electrical properties	
Volume resistivity (<i>ρ</i>)	49.3-49.9 nOhm*m

Mechanical properties

Tensile Strengths							
Hardening	Ultimate MPa (PSI)	Yield MPa (PSI)	Tensile Strength acc. ASTM B209 [KSI]	Yield Strength acc. ASTM B209 [KSI] b.com/search/SpecificMaterial.asp? bassnum=MA5052O O]	195 (28000)	89.6 (13000)	
H32 (http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MA5052H32)	228 (33000)	193 (28000)	31.0 - 38.0	>23.0			
H34 (http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MA5052H34)	262 (38000)	214 (31000)	34.0 - 41.0	>26.0			
H36 (http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MA5052H36)	276 (40000)	241 (35000)	37.0 - 44.0	>29.0			
H38 (http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MA5052H38)	290 (42000)	255 (37000)	>39.0	>32.0			

Uses

Typical applications include marine, aircraft, architecture, general sheet metal work, heat exchangers, fuel lines and tanks, flooring panels, streetlights, appliances, rivets and wire.

The exceptional corrosion resistance of 5052 alloy against seawater and salt spray makes it a primary candidate for the failure-sensitive large marine structures, like tanks of liquefied natural gas tankers.

Weldability

Weldability – Gas: Good

Weldability – Arc: Very Good

Weldability – Resistance: Very Good

Brazability: Acceptable

Solderability: Not recommended

References

1. ASM Handbook, Volume 2: Properties and Selection: Nonferrous Alloys and Special-Purpose Materials ASM Handbook Committee, p 91 DOI: 10.1361/asmhba0001060 (<https://materialsdata.nist.gov/dspace/xmlui/bitstream/handle/11115/179/Properties%20of%20Wrought%20Aluminum.pdf?sequence=3&isAllowed=y>)
2. Asre Sanat Eshragh Co.'s datasheet on 5052 aluminum (<https://web.archive.org/web/20140327122913/http://www.asecoltd.com/userfiles/cliparts/Aluminum%205052.pdf>), accessed 2015-12-21

Aluminium alloy table

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